POLITECNICO MILANO 1863

DEPARTMENT OF AEROSPACE SCIENCE AND TECHNOLOGY

FLY MI

26TH NOVEMBER 2024

PRELIMINARY DESIGN REVIEW

13 - Fly-Mi - Politecnico di Milano

TEAM MEMBERS

Marta Maria Catalano, *BSc Aerospace Engineering, 3rd year*
Filippo Coacci, *MSc Aeronautical Engineering, 2nd year*
Luigi Kowské Ernesto Hayashida, *BSc Aerospace Engineering, 3rd year*
Paola Mansueti, *BSc Aerospace Engineering, 3rd year*
Alex Cristian Turcu, *MSc Space Engineering, 2nd year*
Giorgia Pallara, *MSc Aeronautical Engineering, 2nd year*
Lorenzo Martella, *BSc Mechanical Engineering, 3rd year*
Domenico Lo Faro, *BSc Aerospace Engineering, 3rd year*
Giuseppe Rizi, *MSc Aeronautical Engineering, 1st year*
Luca Baldi, *BSc Aerospace Engineering, 2nd year*

ACADEMIC LEAD

Lorenzo Trainelli

TEAM LEADER ______________ *ACADEMIC LEAD* ______________

BERCELLA · Ansys · DASSAULT SYSTEMES · ELIGIO RE FRASCHINI dal 1946 · xenia Thermoplastic Specialties · TREED · C-I-T composite materials italy · MKS Servo-tech · makita

1 Introduction

On the 26th of November 2024 the Preliminary Design Review (PDR) took place with the main objectives of evaluating the progress and alignment of the project with competition requirements, ensuring adherence to the timeline and identifying any delays or issues that need to be addressed. This review also provides a foundation for redirecting the team's efforts and improving specific activities where necessary. The review was led by the team's academic supervisor and consisted of a presentation of the preliminary design by the team, followed by the professor's feedback. The minutes of the meeting are detailed in the following text. It is worth noting that the team's design presentation has been intentionally summarized, as its technical content mirrors what is already outlined in the concept paper. The supervisor's feedback, along with the planned actions and the designated responsible for their implementation, are then presented as per the prescribed regulations.

2 Project Status

The first point addressed was the overall status of the project from a planning perspective. The project is successfully following the timeline outlined in the Gantt chart, though some activities have fallen slightly behind schedule. These include the computational calculations of the aircraft's characteristics and finalizing the tail and tail-boom design. This delay has pushed the schedule back by approximately one week.

3 Performance Review

Key performance parameters were then reviewed to ensure compliance with competition standards, including:

- **Propeller Selection:** A pool of propellers capable of providing sufficient thrust for take-off within the 10x10m box has been identified. These will be further investigated during flight testing and a possible wind tunnel testing. Should the wind tunnel testing happen, it will take place during the last two weeks of February.
- **Wing Design:** The optimized wing geometry has been finalized. The primary design goals were maximum efficiency in cruise conditions, sufficient lift capabilities for take-off and landing, and desirable stall characteristics. Regarding this last point, a negative wing-tip twist was introduced to ensure a smooth stall and avoid aileron stall.
- **Stability and Control capabilities:** The wing control surfaces were sized to meet the stability and control requirements, and the flap was designed to meet take-off and landing performance needs. Furthermore, a computational model that calculates the plane centre of gravity was developed. The use of the code aims at making sure that the final product is correctly balanced both in flight and taxiing conditions.
- **Payload Release System:** The payload bay and releasing mechanism has been modified in accordance with the resized Micro UAV System.
- **Preliminary Weight Estimation:** A preliminary structure dimensioning was performed using both analytical and computational methods. The first iteration of the various component design allowed for a gross component estimation, which showed that the development is in line with the conceptual design expectations. The estimated payload total mass will be of about 3 kg.
- **Flight Control:** A preliminary phase of testing was successfully completed, allowing to tune the autonomous software parameters. However, additional testing and calibration are needed to improve the accuracy of flight velocity and altitude.
- **V-Tail testing:** A test of the novel V-tail design on a previous model drone is planned in order to improve the understanding of the tail behavior, particularly in windy conditions, and to better understand the autonomous pilot setup.

4 Feedback and Key Discussion Points

The feedback and discussion session with the supervisor focused on two main topics:

- **Testing:** The importance of precision in testing was emphasized, with a particular focus on the sensitivity of stability and controllability measurements to piloting style and repeatability. It was also recommended to produce multiple V-tail geometries for parallel testing to optimize the final design.
- **Risk Mitigation:** The importance of scheduling critical activities, such as testing, construction, and repairs, well in advance to allow for buffer periods for potential delays was emphasized.

5 Conclusions

Based on the review, the Gantt chart has been updated to account for additional time needed for manufacturing spare parts and supporting the testing phases. The team will allocate additional time for computational modeling and testing of the V-tail configuration. This phase is scheduled for the period going from the 01/03/2025 to 23/03/2025 allowing sufficient time for both computational analysis and physical testing. Testing will involve the partecipation of both the Electronics Propulsion and Control and the Flight Mechanics departments. The CTO will oversee the coordination between departments to achieve these objectives effectively.
The overall design approach was commended, noting that it is well-structured and it is progressing in the right direction. The team is confident that the planned corrective actions will ensure the successful completion of the preliminary design phase.